

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

# Chapter 12: Reflection

Previous year board questions, topic-wise: 7 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

## 1 Reflection of figures

1. Plot  $A(1, 3)$ ,  $B(1, 2)$  and  $C(3, 0)$  on graph paper. (a) Reflect  $A$  and  $B$  in the  $x$ -axis to get  $E$  and  $D$ . (b) Reflect  $A$  and  $B$  in the origin to get  $F$  and  $G$ . (c) Reflect  $A$ ,  $B$  and  $C$  in the  $y$ -axis to get  $J$ ,  $I$  and  $H$ . (d) Join  $A, B, C, D, E, F, G, H, I, J$  in order and name the closed figure. **ICSE 2025, 5 marks**
2. Plot  $A(0, 3)$ ,  $B(2, 1)$  and  $C(4, -1)$ . (a) Reflect  $B$  and  $C$  in the  $y$ -axis to get  $B'$  and  $C'$ , and write their coordinates. (b) Reflect  $A$  in the line  $BB'$  to get  $A'$ , and write its coordinates. (c) Join  $A, B, A', B'$  in order and name the figure. **ICSE 2024, 5 marks**
3. Plot  $A(0, 4)$ ,  $B(1, 1)$  and  $C(4, 0)$ . (a) Reflect  $A$  and  $B$  in the  $x$ -axis to get  $E$  and  $D$ . (b) Reflect  $B$  in the origin to get  $F$ , and write its coordinates. (c) Reflect  $B$  and  $C$  in the  $y$ -axis to get  $H$  and  $G$ . (d) Join  $A, B, C, D, E, F, G, H, A$  in order and name the closed figure. **ICSE 2023, 5 marks**
4. Plot  $A(-4, 0)$ ,  $B(-3, 2)$ ,  $C(0, 4)$ ,  $D(4, 1)$  and  $E(7, 3)$ . Reflect  $B, C, D$  and  $E$  in the  $x$ -axis to get  $B', C', D'$  and  $E'$ . Join  $A, B, C, D, E, E', D', C', B', A$  in order and name the closed figure. **ICSE 2020, 4 marks**
5. Plot  $A(0, 5)$ ,  $B(3, 0)$ ,  $C(1, 0)$  and  $D(1, -5)$ . Reflect  $B, C$  and  $D$  in the  $y$ -axis to get  $B', C'$  and  $D'$ , and write their coordinates. Join  $A, B, C, D, D', C', B', A$  in order and name the closed figure. **ICSE 2019, 4 marks**
6. Plot  $A(0, 4)$ ,  $B(2, 3)$ ,  $C(1, 1)$  and  $D(2, 0)$ . Reflect  $B, C$  and  $D$  in the  $y$ -axis to get  $B', C'$  and  $D'$ , and write their coordinates. Join  $A, B, C, D, D', C', B', A$  in order and write the equation of the line of symmetry of the figure. **ICSE 2017, 5 marks**

## 2 Invariant points

7.  $ABCD$  is a quadrilateral with vertices  $A(2, 2)$ ,  $B(2, -2)$ ,  $C(0, -1)$  and  $D(0, 1)$ . (a) Reflect  $ABCD$  in the  $y$ -axis to get  $A'B'CD$ , and write the coordinates of  $A'$  and  $B'$ . (b) Name two points that are invariant under this reflection. (c) Name the polygon  $A'B'CD$ . **ICSE 2018, 4 marks**

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.